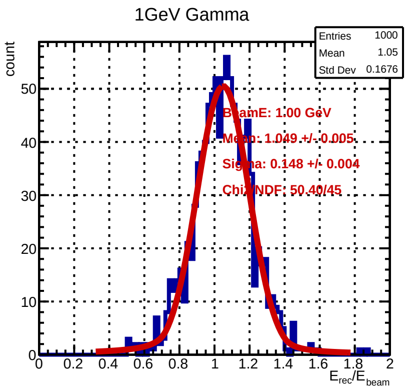
Different Kinds of Energy Regression

- Fitting Functions
  - Linear function : $E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL}$
  - Linear function w/ ratio : $E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL} + p3 * E_{HCAL}/(E_{ECAL}+E_{HCAL})$
  - Quadratic function : $E_{rec} = p0 + p1 * E^2_{ECAL} + p2 * E^2_{HCAL}$

- Weighting (Events enter fitting with a certain weight)
  - Equal weighting : weighting = 1 (all the events are equal)
  - Square-root of beam energy : weighting = $\sqrt{E_{beam}}$ (higher energy beam has more weight)



Linear energy regression

$$E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL}$$

Fit E_{rec} with double-side crystal ball function

- Energy resolution = σ / E_{beam} (%)
- Energy bias = μ / E_{beam}